

Chapter 8

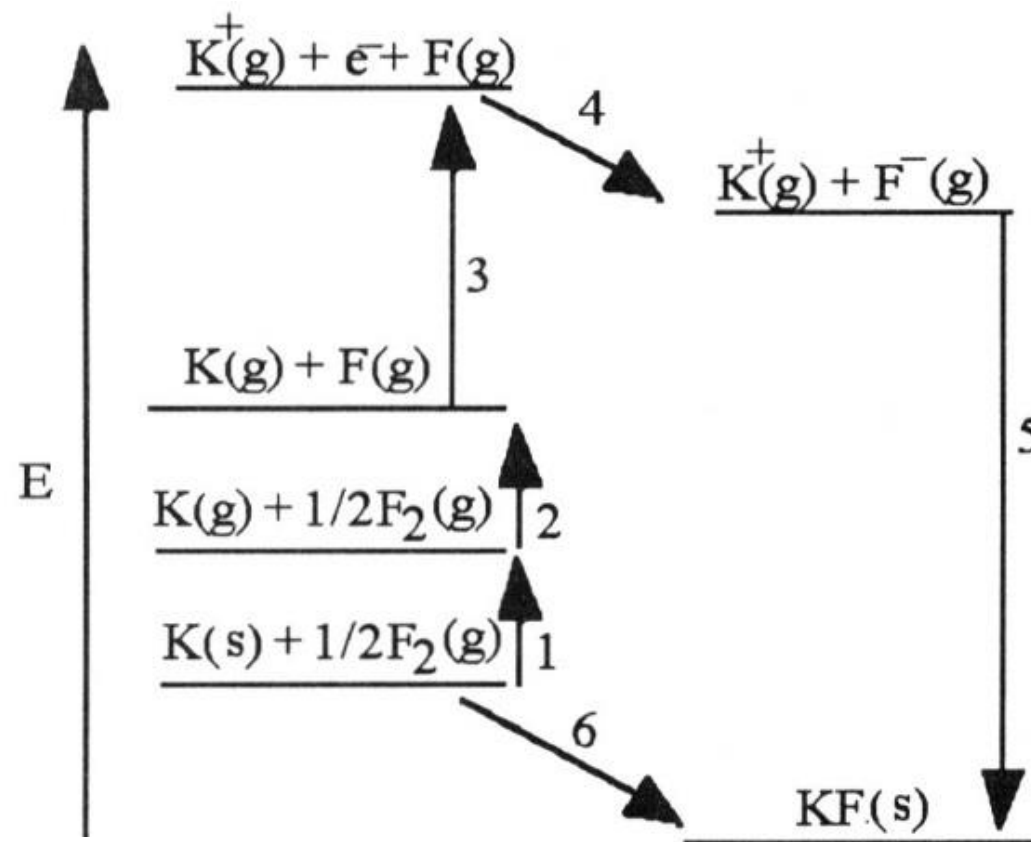
Basic Concepts of Chemical Bonding

1. In ionic bond formation, the lattice energy of ions _____ as the magnitude of the ion charges _____ and the radii _____.

- A) increases, decrease, increase
- B) increases, increase, increase
- C) decreases, increase, increase
- D) increases, increase, decrease
- E) increases, decrease, decrease

2. Which energy change corresponds to the electron affinity of fluorine
 C ?

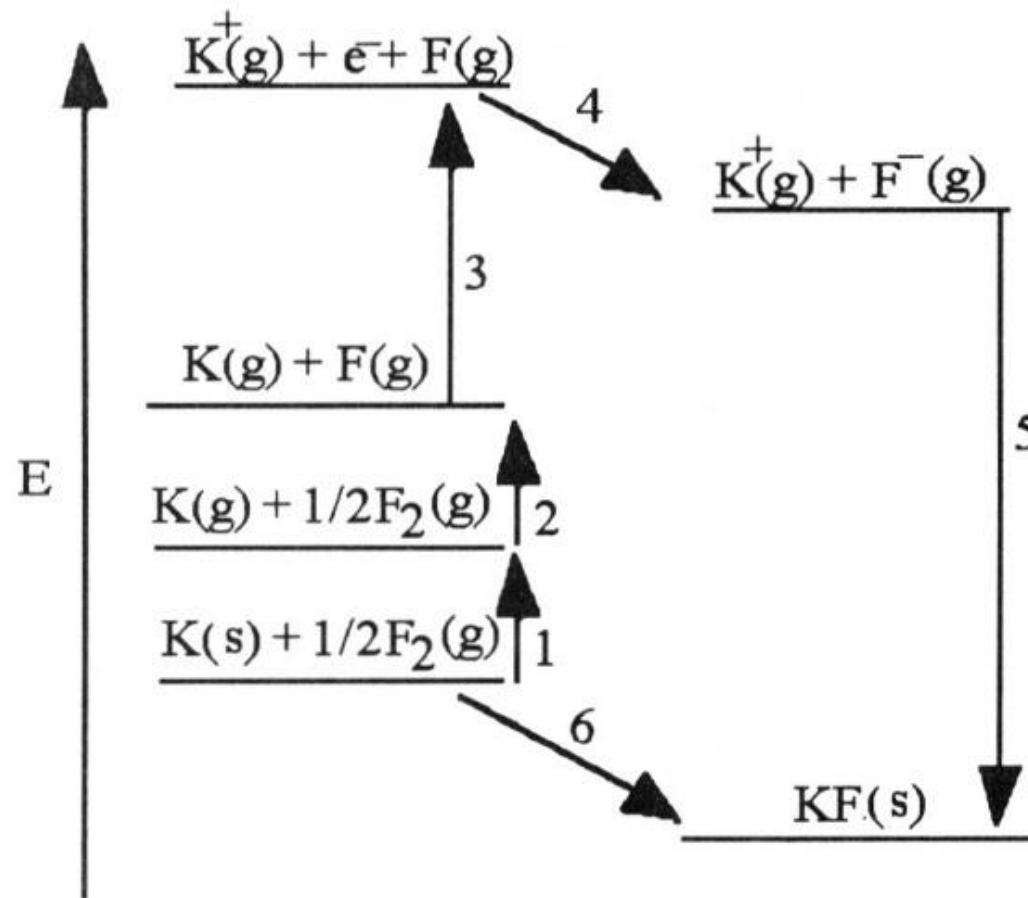
- A) 2
- B) 5
- C) 4
- D) 1
- E) 6



Electron affinity: the energy change accompanying the addition of an electron to
a gaseous atom, $\text{Cl(g)} + \text{e}^- \longrightarrow \text{Cl}^-\text{(g)}$

3. Which energy change corresponds to the first ionization energy of potassium D?

- A) 2
- B) 5
- C) 4
- D) 3
- E) 6



4. The type of compound that is most likely to contain a covalent bond is C.

A) one that is composed of a metal from the far left of the periodic table and a nonmetal from the far right of the periodic table

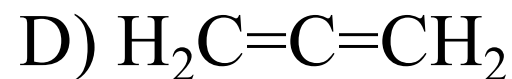
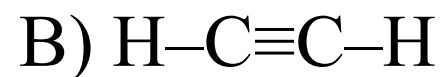
B) a solid metal

C) one that is composed of only nonmetals

D) held together by the electrostatic forces between oppositely charged ions

E) There is no general rule to predict covalency in bonds

5. In which of the molecules below is the carbon-carbon distance the shortest B?



6. Of the molecules below, the bond in D is the most polar. Polarity $\propto \Delta EN$

- A) HBr
- B) HI
- C) HCl
- D) HF
- E) H₂

Electronegativity: the ability of an atom in a molecule to attract electrons to itself.

Trends:

- 1) Electronegativity increase from left to right across a period;
- 2) Electronegativity decrease with increasing atomic number in a group;

7. The bond length in an HI molecule is 1.61 Å and the measured dipole moment is 0.44 D. What is the magnitude (in units of e) of the negative charge on I in HI B?
(1 debye = 3.34×10^{-30} coulomb-meters; $e = 1.6 \times 10^{-19}$ coulombs)

$$\mu = Qr$$

$$0.44 \times 3.34 \times 10^{-30} = Q (1.6 \times 10^{-19}) (1.61 \times 10^{-10})$$
$$Q = 0.057$$

- A) 1.6×10^{-19}
- B) 0.057
- C) 9.1
- D) 1
- E) 0.22

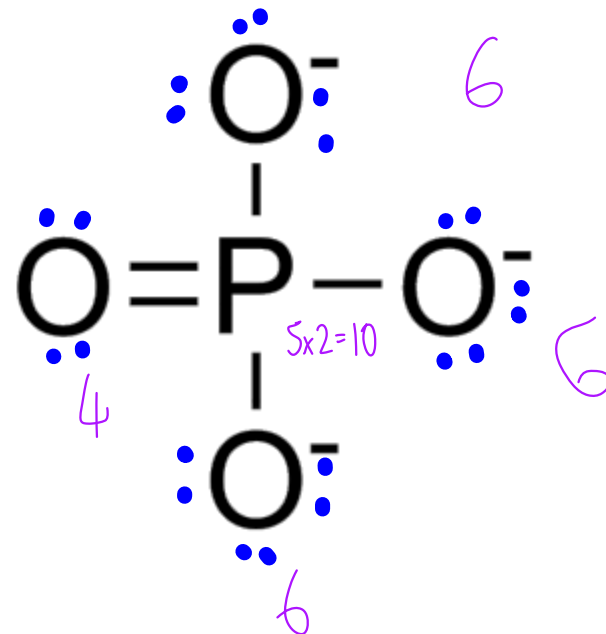
8. The ion PO_4^{3-} has E valence electrons.

$$5 + 6 \times 4 + 3 = 32$$

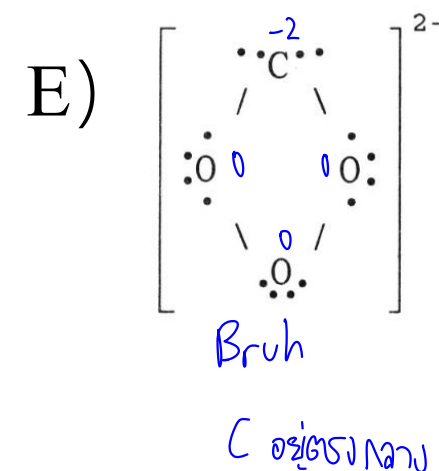
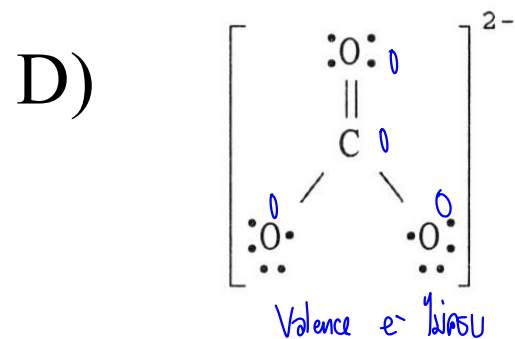
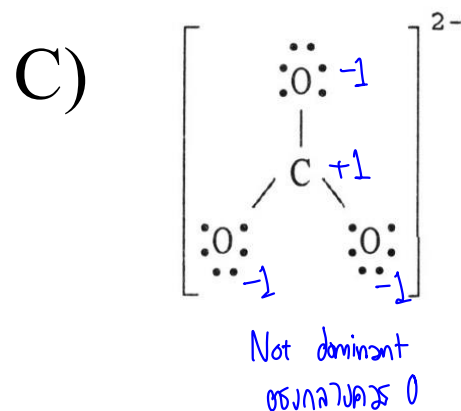
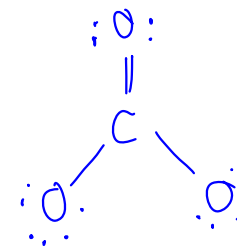
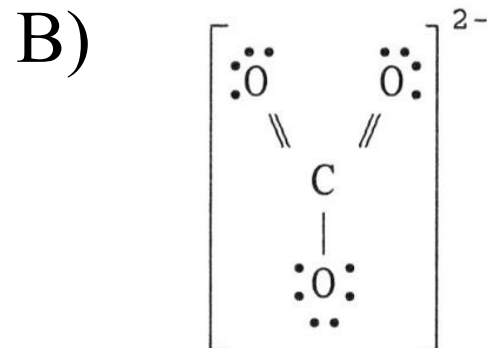
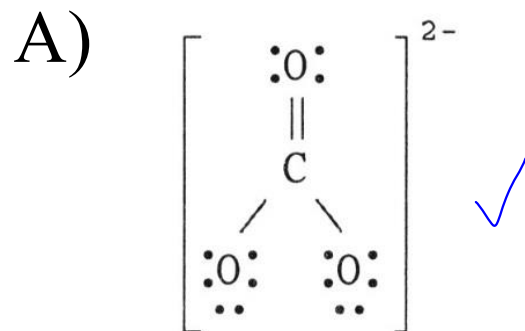
22 nonbonding

10 bonding

- A) 14
- B) 24
- C) 27
- D) 29
- E) 32

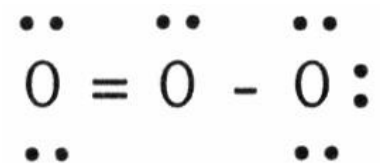


9. The Lewis structure of the CO_3^{2-} ion is A.



$$\text{Formal charge} = \text{valence electrons} - \frac{1}{2}(\text{bonding electrons}) - \text{nonbonding electrons}$$

10. In the resonance form of ozone shown below, the formal charge on the central oxygen atom is B .



- A) 0
- B) +1
- C) -1
- D) +2
- E) -2

11. To convert from one resonance structure to another,
C.

- A) only atoms can be moved
- B) electrons and atoms can both be moved
- C) only electrons can be moved
- D) neither electrons nor atoms can be moved
- E) electrons must be added

12. As the number of covalent bonds between two atoms increases, the distance between the atoms _____ and the strength of the bond between them D.

- A) increases, increases
- B) decreases, decreases
- C) increases, decreases
- D) decreases, increases
- E) is unpredictable

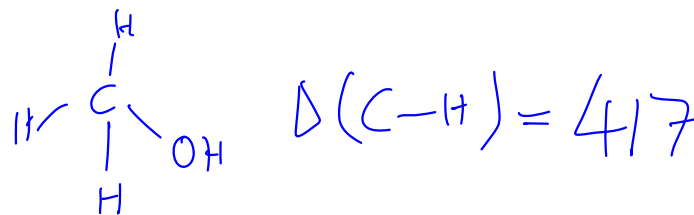
13. The reaction below is used to produce methanol:



(a) Calculate the C-H bond energy given the following data:

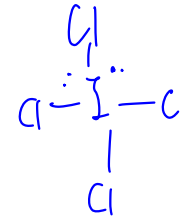
Bond	D (kJ/mol)
C≡O	1072
H—H	436
C—O	358
O—H	463

$$-128 = 1072 + 872 - 3(\text{C-H}) - 358 - 463$$



(b) The tabulated value of the (C-H) bond energy is 413 kJ/mol. Explain why there is a difference between the number you have calculated in (a) and the tabulated value.

↳ Avg. bond energy from this combo in many molecules



Answer:

(a)

$$\Delta H_{\text{rxn}} = D(\text{C}\equiv\text{O}) + 2 D(\text{H}-\text{H}) - [3 D(\text{C}-\text{H}) + D(\text{C}-\text{O}) + D(\text{O}-\text{H})]$$

$$3 D(\text{C}-\text{H}) = -\Delta H_{\text{rxn}} + D(\text{C}\equiv\text{O}) + 2 D(\text{H}-\text{H}) - D(\text{C}-\text{O}) - D(\text{O}-\text{H})$$

$$D(\text{C}-\text{H}) = (128 + 1072 + 2(436) - 358 - 463)/3 = 417$$

$$D(\text{C}-\text{H}) = 417 \text{ kJ/mol}$$

Note: Here the energy change regarding phase change of CH_3OH is ignored. Though the enthalpy of the reaction, ΔH_{rxn} , can be estimated as follows:
 $\Delta H_{\text{rxn}} = \Sigma(\text{bond enthalpies of bonds broken}) - \Sigma(\text{bond enthalpies of bonds formed})$,
 this kind of estimation is only applied for the gas-phase reactions.

(b) Tabulated values, like those in Table 8.4, are averaged from many bond energies measured for C-H bonds in many different molecules.